

GROVE, Australia

WMO: 959770

Lat: 42.986S

Lon: 147.074E

Elev: 65

StdP: 100.55

Time Zone: 10.00 (AUS)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-1.6	-0.6	-1.4	3.4	11.1	-0.3	3.7	8.0	9.9	10.2	8.4	9.7	0.8	30	0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.3	28.7	17.4	26.2	16.9	24.0	16.3	19.6	24.4	18.3	23.7	17.3	22.1	4.8	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.5	13.4	20.0	16.7	12.0	19.4	15.5	11.1	18.6	56.4	24.2	52.1	24.0	48.8	22.3	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	7.6	6.8	-3.0	35.7	0.9	1.9	-3.7	37.0	-4.2	38.1	-4.7	39.2	-5.4	40.5		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	15.9	15.5	14.3	11.6	9.5	7.3	7.0	7.9	9.5	11.0	12.7	14.2
	DBStd	4.23	3.38	3.13	3.22	2.99	2.90	2.60	2.46	2.74	2.76	3.02	3.04	3.00
	HDD10.0	404	1	1	3	16	46	90	99	77	41	22	7	2
	HDD18.3	2594	93	89	132	204	274	332	351	322	264	227	173	133
	CDD10.0	893	182	156	136	63	29	7	7	14	27	54	87	133
	CDD18.3	43	16	11	7	1	0	0	0	0	1	1	2	5
	CDH23.3	637	228	135	95	10	0	0	0	0	2	18	56	94
	CDH26.7	187	77	43	25	1	0	0	0	0	0	3	12	27
(14) Wind	WSAvg	2.6	3.3	2.7	2.6	2.2	2.3	1.8	2.1	2.5	2.9	3.1	3.0	3.2
(15) Precipitation	PrecAvg	743	50	47	50	62	58	55	76	78	71	69	65	61
	PrecMax	1099	146	162	147	338	158	125	223	172	175	182	143	180
	PrecMin	498	9	5	8	6	17	9	27	24	25	32	6	5
	PrecStd	134	33	40	32	54	35	26	42	41	36	35	31	41
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.0	31.9	29.5	24.5	20.0	16.6	15.8	18.6	21.5	25.8	28.6	29.7
		MCWB	19.1	18.7	17.7	16.9	12.6	12.1	10.0	11.4	12.7	14.5	16.0	17.0
	2%	DB	28.6	27.1	25.8	21.9	17.9	15.2	14.4	16.2	18.5	22.3	24.8	25.9
		MCWB	17.9	17.3	17.0	14.5	12.4	10.7	9.5	10.4	11.9	13.2	16.7	16.6
	5%	DB	25.6	24.2	23.1	19.4	16.5	14.0	13.2	14.7	16.8	19.7	21.5	22.9
		MCWB	17.0	17.0	17.1	14.1	11.6	9.9	9.0	9.7	11.2	12.8	14.9	15.6
	10%	DB	23.1	22.2	20.7	17.5	15.1	12.8	12.1	13.3	15.1	17.4	19.2	20.7
		MCWB	15.8	15.8	15.3	13.0	10.9	9.3	8.5	9.2	10.2	11.7	13.8	14.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.0	20.2	22.9	19.0	14.8	13.9	11.5	12.4	16.4	16.5	19.4	19.2
		MCDB	25.7	25.9	23.3	20.3	17.0	15.3	14.0	15.5	17.0	20.6	21.5	24.9
	2%	WB	19.3	18.8	19.1	16.0	13.1	11.5	10.2	11.4	13.2	14.9	17.4	17.8
		MCDB	26.1	25.1	23.1	20.5	16.4	14.0	13.3	15.0	17.2	20.4	22.2	22.8
	5%	WB	17.7	17.6	17.3	14.5	12.2	10.5	9.4	10.4	11.7	13.3	15.9	16.6
		MCDB	23.4	23.4	21.8	18.1	15.7	13.2	12.6	14.1	15.4	18.3	20.6	21.1
	10%	WB	16.5	16.5	15.9	13.3	11.3	9.7	8.8	9.4	10.6	12.0	14.4	15.4
		MCDB	21.7	21.1	20.3	16.9	14.6	12.5	11.8	12.9	14.5	16.7	18.7	20.1
(35) Mean Daily Temperature Range	5% DB	MDBR	12.3	11.9	11.2	10.7	9.2	9.2	8.8	9.2	9.8	10.7	10.8	11.1
		MCDBR	17.7	16.2	15.4	14.2	10.6	10.5	9.6	10.9	12.5	15.2	16.0	15.7
		MCWBR	8.2	7.8	8.5	8.2	6.6	6.1	5.6	5.9	7.2	8.0	8.5	7.9
	5% WB	MCDBR	14.9	14.2	13.6	12.8	10.0	9.0	8.3	9.6	11.0	13.4	14.1	13.4
		MCWBR	8.5	7.6	9.0	8.4	6.8	6.1	5.5	6.1	7.6	7.9	8.8	8.2
(40) Clear-Sky Solar Irradiance	taub		0.346	0.333	0.333	0.315	0.307	0.294	0.296	0.309	0.332	0.336	0.335	0.340
	taud		2.509	2.562	2.555	2.585	2.588	2.604	2.603	2.569	2.490	2.494	2.517	2.515
	Ebn at Noon		979	965	918	864	798	776	804	858	902	951	983	991
	Edh at Noon		110	99	90	75	63	56	60	74	95	105	108	111
(44) All-Sky Solar Radiation	RadAvg		6.39	5.47	4.03	2.73	1.66	1.32	1.52	2.27	3.42	4.83	5.80	6.40
	RadStd		0.47	0.34	0.23	0.16	0.11	0.11	0.08	0.16	0.17	0.27	0.45	0.38

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (27 neighbors)	+0.32	N/A	N/A	+0.68	N/A	N/A	N/A	-100	+106	+14

Nomenclature: See separate page